

Embeddings · Similarity Search · Scale

Keyword search sees them as strangers. Embeddings see them as cousins. One API call gives you the semantic fingerprint that powers RAG, recommendation, clustering, and classification.

Which two terms in YOUR domain mean the same thing but don't share letters?

WHAT'S INSIDE

- 01** Embeddings — dense fingerprints
- 02** ANN algorithms
- 03** 2026 vector-DB landscape
- 04** Metadata filters + hybrid search
- 05** Analogy

WHY THIS LESSON MATTERS

"Bengaluru techie" and "Bangalore developer" — same meaning. Same vector.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Embeddings as semantic fingerprints.

2

ANN algorithms (HNSW, IVF) conceptually.

3

Compare ChromaDB, Pinecone, Weaviate, pgvector, Qdrant.

4

Metadata filters + hybrid search.

5

Pick a vector DB by scale + hosting constraint.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Embeddings — dense...

Map anything to a 384–3072 dim
vector

02

ANN algorithms

Key concept of this lesson.

03

2026 vector-DB landscape

< 1 M + dev → Chroma. 1 M–100 M +
zero ops → Pinecone. On Postgres →
pgvector.

04

Metadata filters + hybrid...

Real queries combine semantics +
hard filters

01

PART 1 OF 4

Embeddings — dense...

Map anything to a 384-3072 dim vector

CONCEPT 1 OF 4

Embeddings — dense fingerprints

Same meaning → nearby vectors.

Cosine similarity scores closeness.

Works on text, images, audio, code.

text-embedding-3-small: 1536 dims, ₹0.02 per 1M tokens.

EXAMPLE

1536 dim

The "shape" of a concept in compressed form.

DEFINITION

Embeddings — dense fingerprints

Map anything to a 384–3072 dim vector

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

ANN algorithms

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
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2026 vector-DB landscape

< 1 M + dev → Chroma. 1 M–100 M + zero ops → Pinecone. On Postgres → pgvector.

DEFINITION

2026 vector-DB landscape

< 1 M + dev → Chroma. 1 M–100 M + zero ops → Pinecone. On Postgres → pgvector.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Metadata filters + hybrid search

`query_embeddings=[q_emb], where={"country":"India"}`

Dense (semantic) + sparse (BM25) = hybrid.

Hybrid often +10% recall.

Don't ship semantic-only production search.

PATTERN

Dense + Sparse

Sparse rescues exact-match terms (SKU codes, IDs).

KEY POINTS

- `query_embeddings=[q_emb], where={"country":"India"}`
- Dense (semantic) + sparse (BM25) = hybrid
- Hybrid often +10% recall
- Don't ship semantic-only production search

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where will you apply this in your next project?

Q2

What was the single biggest 'aha' from this lesson?

Q3

Which habit will you start using this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Fastest ANN for most needs:

A

Brute force

B

HNSW

C

Binary search

D

Hash

Fastest ANN for most needs:



CORRECT ANSWER

HNSW

WHY

HNSW scales to 100M+ vectors with <10ms latency

pgvector sweet spot:

A

Team on Postgres already

B

Always

C

Never

D

Images only

pgvector sweet spot:



CORRECT ANSWER

Team on Postgres already

WHY

Zero-ops extension when you already run Postgres

Hybrid search combines:

A

Two models

B

Dense + BM25

C

SQL + Mongo

D

None

Hybrid search combines:



CORRECT ANSWER

Dense + BM25

WHY

Dense vectors + sparse keyword search unioned + reranked

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Same meaning → nearby vectors.
Cosine similarity scores closeness.
Works on text, images, audio, code.
text-embedding-3-small: 1536 dims, ₹0.02 per 1M tokens.

EXAMPLE

1536 dim

The "shape" of a concept in compressed form.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1

Embeddings — dense...

2

ANN algorithms

3

2026 vector-DB landscape

4

Metadata filters + hybrid...

The vector DB is a search index.

"Everything you already know about search indexes still applies."

DELIVERABLES

- NEXT
- Lesson 5 — Building AI Agents.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 5

What you can now do

Each one of these is now part of your working vocabulary.

1

Pick the right tool — never the loudest one.

2

Practice the framework on a real example.

3

Watch for the three classic pitfalls.

4

Document your decision in one sentence.

5

Carry the discipline forward to the next lesson.